## Measuring Inflation

Before we introduce inflation into an engineering problem, we need to measure its effect. Generally, inflation is measured in percentage term which is simply obtained by calculating the percentage change in current price index over the previous year.

Price indices are developed on the basis of market prices of various goods & services under the study and it is used for measuring inflation.

The most commonly used indices are:

A. Consumer Price Index (CPI)
B. Wholesale Price Index (WPI)
C. GDP deflator

### A. Consumer Price Index (CPI)

CPI is an index which is used to measure of overall change in cost of good/services purchased by typical consumers like urban consumers. It is a way of finding the cost of living (housing, food & beverages, transportation, entertainment, education, medical care, other goods & services).

**How to calculate CPI?**

**Step 1; Find the basket of goods & services**

Let, a basket of consumer goods contains 4 breads & 2 eggs.

**Step 2; Find the prices of different years**

| Year | Price of bread/unit | Price of egg/unit |
|---|---|---|
| 2015 | Rs. 1 | Rs. 2 |
| 2016 | Rs. 2 | Rs. 3 |
| 2017 | Rs. 3 | Rs. 4 |

**Step 3; Compute the actual cost of basket for each year**

2015; Rs. $1\times$ 4Breads + Rs.$2 \times 2$ Eggs = Rs. 8

2016; Rs. $2\times$ 4Breads + Rs. $3 \times 2$ Eggs = Rs. 14

2017; Rs. $3\times$ 4Breads + Rs.$4 \times 2$ Eggs = Rs. 20

**Step 4;** Choose a base year (2015) and compute CPI by using formula; $\frac{P_1}{P_0} \times 100$

CPI of 2015; Rs. $\frac{8}{8} \times 100 = 100$

CPI of 2016; Rs. $\frac{14}{8} \times 100 = 175$

CPI of 2017; Rs. $\frac{20}{8} \times 100 = 250$

*Where,* $P_1$ = *Curent price*

$P_0$ = *Base year price*

**Step 5;** Use CPI to compute the inflation rate

$$\text{Rate of inflation} = \frac{CPI_2 - CPI_1}{CPI_1} \times 100$$

$$\text{or, Rate of inflation} = \frac{\text{Change in CPI}}{\text{Previous year's CPI}} \times 100$$

$$\therefore \text{ Inflation in 2016; } \frac{175 - 100}{100} \times 100 = 75\% \text{ as compare to 2015}$$

$$\therefore \text{ Inflation in 2017; } \frac{250 - 175}{175} \times 100 = 43\% \text{ as compare to 2016}$$

## B. Wholesale Price Index (WPI)

It is an index which is used to measure the average price changes of goods/services between years (times) at the level of wholesale or big sale prices. It is a measure of changes in prices charged by manufactures or wholesalers so it is also known as Producer Price Index (PPI).
The process of constructing WPI or PPI is similar to that of CPI.

## C. GDP Deflator

GDP deflator is a price index which measures the changes in overall prices of final goods & services. It is used as an indicator of average price changes in the economy, so it measures inflation rate over the country.
"The ratio of nominal GDP to real GDP times 100 is called GDP deflator."

$$\text{GDP deflator} = \frac{\text{Nominal (Current Price)GDP}}{\text{Real (Base Year Price) GDP}} \times 100$$

*[Note: **Nominal** = current output x current price & **Real** = current output x base year price]*

### How to calculate GDP deflator?

**Step 1;** Find the goods produced in the economy and their prices of different years.
Let economy produces two goods – Bread & Egg.

| Year | Price of Bread | Quantity of Bread | Price of Egg | Quantity of Egg |
|---|---|---|---|---|
| 2015 | Rs. 1 | 100 | Rs. 2 | 50 |
| 2016 | Rs. 2 | 150 | Rs. 3 | 100 |
| 2017 | Rs. 3 | 200 | Rs. 4 | 150 |

**Step 2; Calculate Nominal GDP**

2015; Rs. $1 \times 100$ Bread + Rs.$2 \times 50$ Eggs = 100 + 100 = Rs. 200

2016; Rs. $2 \times 150$ Bread + Rs.$3 \times 100$ Eggs = 300 + 300 = Rs. 600

2017; Rs. $3 \times 200$ Bread + Rs.$4 \times 150$ Eggs = 600 + 600 = Rs. 1200

**Step 3; Calculate Real GDP (Assuming Base Year 2008)**

2015; Rs. $1 \times 100$ Bread + Rs. $2 \times 50$ Eggs = Rs. 200

2016; Rs. $1 \times 150$ Bread + Rs. $2 \times 100$ Eggs = Rs. 350

2017; Rs. $1 \times 200$ Bread + Rs. $2 \times 150$ Eggs = Rs. 500

**Step 4; Calculate GDP Deflator;** $\frac{\text{Nominal (Current Price)GDP}}{\text{Real (Base Year Price) GDP}} \times 100$

GDP deflator in 2015; Rs. $\frac{200}{200} \times 100 = 100$

GDP deflator in 2016; Rs. $\frac{600}{350} \times 100 = 171$

GDP deflator in 2017; Rs. $\frac{1200}{500} \times 100 = 240$

**Step 5; Calculate Rate of Inflation by using formula:**

$$\frac{\text{GDP deflator in current year} - \text{GDP deflator of previous year}}{\text{GDP deflator of privious year}} \times 100$$

or, $= \frac{\text{Change in GDP deflator}}{\text{Previous year's GDP deflator}} \times 100$

$\therefore$ Rate of inflation in 2016 $= \frac{171 - 100}{100} \times 100 = 71\%$ as compare to 2015.

$\therefore$ Rate of inflation in 2017 $= \frac{240 - 171}{171} \times 100 = 40.35\%$ as compare to 2016.

| *Note: CPI is the most frequently used measure of prices in the world* |
|---|

**Example: 8.1**

If an index (CPI) moves from 200 to 216 in a year, find the rate of change in price or inflation.

***Solution:***

When CPI is given, the rate of inflation would be,

$$\frac{CPI_2 - CPI_1}{CPI_1} \times 100$$

$$= \frac{216 - 200}{200} \times 100 = 8\% \text{ within a year}$$

**Example: 8.2**

Calculate average (compound) inflation rate when CPI has risen from 176 to 216 over the past 3 years.

***Solution:***

Given; P=216, N = 3 and FW = 216

We know, $F = P(1+f)^N$ *[where f = rate of inflation]*

The average/compound inflation rate (f) can be calculated as,

$176(1+f)^3 = 216$

or, $(1+f)^3 = \frac{216}{176} = 1.2273$

or, $(1+f) = (1.2273)^{1/3} = 1.071$

or, $f = 0.071$ or $7.1\%$

$\therefore$ $f = 7.1\%$ per year

**Example: 8.3**

Calculate rate of inflation for each year & average inflation over the years from the following information.

| Year | Cost (Rs) |
|---|---|
| 0 | 5,04,000 |
| 1 | 5,38,400 |
| 2 | 5,77,000 |
| 3 | 6,29,500 |

**Solution:**

Inflation rate during year 1;

$$f_1 = \frac{5,38,400 - 5,04,000}{5,04,000} \times 100$$

$$= 6.83\%$$

Inflation rate during year 2;

$$f_2 = \frac{5,77,000 - 5,38,400}{5,38,400} \times 100$$

$$= 7.17\%$$

Inflation rate during year 3;

$$f_3 = \frac{6,29,500 - 5,77,000}{5,77,000} \times 100$$

$$= 9.10\%$$

Average inflation rate (f) over the years;

We know,

$$F = P(1 + f)^N$$

or, $6,29,500 = 5,04,000\,(1 + f)^3$

or, $(1 + f)^3 = 1.249$

or, $1 + f = 1.0769$

or, $f = 0.0769 = 7.69\%$

$\therefore$ Average inflation rate (f) = 7.69% per year

To introduce the effect of inflation in economic analysis, we need to define following inflation related terms.

- **Constant (real) cash flow ($A'_n$)**

  It is the cash flow free from the effect of inflation or deflation. It eliminates inflationary effect by converting all cash flows to base year that have constant purchasing power and it is called constant or real cash flow (dollar).

  Constant cash flow represents the amount expressed in terms of purchasing power of the base year. It is denoted by $A'_n$.

- **Actual (current) cash flow ($A_n$)**

  It is the cash flow (paid) at the time of purchasing goods & services. It is the estimated future cash flows by adjusting/anticipating/expecting inflationary or deflationary effects. It is denoted by $A_n$.

- **Conserves from constant to actual cash flow**

  We can determine equivalence actual cash flow in year 'n' from the given constant cash flow by using general inflation rate with this formula;

  $$A_n = A'_n (1 + f)^N$$

- **Conversion from actual to constant cash flow:**

  From the above relation, we can write

  $$A'_n = \frac{A_n}{(1 + f)^N}$$